Melvin Shajee

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SUMMARY

Data Analyst and MS IT & Analytics graduate with 3+ years of analytical and data pipelining experience. Proficient in scripting languages, statistics packages, SQL, Python (Pandas, NumPy), R, and BI platforms (PowerBI, Tableau, Lightdash). Track record of migrating legacy dashboards, rules-based automation, and transforming raw datasets into actionable insights.

WORK EXPERIENCE

UPS **Mahwah, NJ**

Data Modeling Co-op *Aug 2026 - Present*

- Proactively looked for process optimization opportunities and designed multiple python scripts to increase workflow automation during the modeling process
- Created automated regression test suite using Powershell and SCAPI
- Modernized internal reporting page using PHP and worked with infrastructure team on deployment
- Created agent to assign classwords according to company standards guide as per AI enablement efforts
- Participated in sprint planning, backlog refinement, project delivery timelines in Azure DevOps

Data Modeling Intern *June 2026 - Aug 2026*

- Ingested spreadsheet data to design foundational ERDs
- Developed logical and physical data models using erwin Data Modeler
- Modeled business rules and business logic within data warehouses
- Apply normalization techniques up to Third Normal Form (3NF) to eliminate data redundancy
- Created data model for HR information from ServiceNow
- Created modern internal reporting webpage for publishing data model reports
- Designed anomaly detection machine learning invoice review tool to minimize billing errors using HTML, CSS, JS, and Python (FastAPI, Uvicorn, pandas, numpy, scikit-learn, and joblib)
- Worked with internal business teams on projects involving Azure exports and GCP

Corcentric **Cherry Hill, NJ**

Business Intelligence Intern *Nov. 2025 - June 2026*

- Migrated customer facing expense, procurement, and aggregate spend analytics dashboards from Sisense to Lightdash while maintaining reporting standards
- Performed quality assurance for existing dashboards monitoring financial performance and financial impact
- Created custom analytics and curated reporting tables for clients using PostgreSQL, dbt, pandas, and Lightdash
- Documented legacy Qlik scripts and migrated currency conversion system to Amazon Redshift
- Automated monthly data refresh and ETL using Python scripting, saving over 4 hours of developer time weekly
- Regularly communicated with manager on task prioritization

Americord Registry **Somerset, NJ**

Lab Technician *June 2025 - July 2025*

- Accessioned more than 10 shipments per day, assigned ISBT number, uploaded data collection sheets and documentation to Salesforce, and assembled kits for cord blood and tissue processing
- Processed cord blood by adding hetastarch, separating the plasma using expressors and centrifuge, and transferring plasma to cryopreservation tubes as part of healthcare research
- Removed blood vessels from umbilical cord, and separated the chorion and amnion from the placenta, added cryoprotectant, and initiated freezing process
- Assembled collection kits, created LPH and VES solutions to decontaminate biosafety cabinets, monitored LN2 tank levels, filled daily logs for temperature and humidity

NJIT Brain Stimulation Lab **Newark, NJ**

Research Assistant *May 2024 - July 2025*

- Affixed EEG caps and recorded data from ferrets before and after spinal injuries as part of biomedical research
- Applied finite impulse response filtering to remove linear trends and minimize artifacts from line noise
- Utilized reference electrode standardization technique to translate multichannel data into a working dataset
- Employed low-pass filter and resampled to prevent aliasing during data processing

- Topographically mapped 5 GB EEG datasets and rejected artifacts using Independent Component Analysis
- Automated data analysis workflows using MATLAB scripting to reduce manual processing time by 40%
- Designed and documented analysis pipelines using the MNE Python library and EEGLAB for analyzing microstates in schizophrenia patients

Kessler Institute **West Orange, NJ**
Research Intern *Aug. 2024 - Dec. 2024*

- Handled missing and incomplete data points from TBI victim gait data using ProtoKinetics Movement Analysis Software
- Programmatically filtered and refined exported data using MATLAB and statistical knowledge
- Visualized key statistics (step length, swing time, stance time, initial double support time, terminal double support time, temporal and spatial symmetry) for research paper submission using Excel
- Conducted literature review assessing the effectiveness of VR in stroke rehabilitation
- Collaborated with NeuroTechR3 R&D team to test and provide feedback on beta stroke rehabilitation software

Spenego Software **Exton, PA**
Quality Assurance Intern *Jun. 2023 – Sept. 2023*

- Executed backtesting simulations within virtualized Linux environments for isolated and reproducible results
- Reported software bugs including steps to reproduce, expected behavior, observed behavior, and logs/error codes in RedMine Wiki
- Identified more than 30 software bugs and usability improvements by collaborating with the development team

PROJECTS

RAG Literature Review Assistant Large Language and Vision Models Final Project

- Created an end to end research agent that uses dense and sparse retrieval, HyDE, Reciprocal Rank Fusion, and the ClaudeAPI to generate clean, citation-ready json for researchers.
- Loaded snapshot of arXiv research paper data
- Create an index using SBERT for dense and BM25 for sparse.
- Used the Claude API for performing HyDE expansion
- Hybrid search using Cosine and BM25 RRF fusion
- Called Claude API to generate justification for suggestion
- SerpAPI for providing links and metadata like citation counts
- https://colab.research.google.com/drive/1tZf0TPKTplNA2v_BcUl6FvZHk8-kL8dA?usp=sharing

ML NY Taxi Trip Duration Prediction Algorithmic Machine Learning Final Project

- Predicted NY Taxi Trip durations using neural networks, ridge regression, XGBoost, SVM, linear regression, and random forest algorithms.
- All techniques were implemented in a Jupyter notebook.
- Key packages were keras, pandas, numpy, and scikit-learn.
- https://github.com/ms358/AlgorithmicMachineLearningProject

Health, Wealth, Education, and Liberty in 200 countries Data Analysis and Visualization Final Project

- Pulled data from worldbank on gender statistics.
- Loaded data into Microsoft PowerBI.
- Used PowerQuery to manage pivots, remove duplicates, and define nulls.
- Organized data into dimension and fact tables as per star schema.
- Created dynamic dashboards to analyze the relationships between the four factors

Respiratory Disease Trend Analysis Data Analysis and Visualization Project

- Pulled data from CDC website on disease incidence rates.
- Loaded data into Tableau
- Joined tables and plotted trends along with forecasted results to tell a data story
- Created maps in order to contextualize the crude rates of respiratory diseases

ML Concussion Evaluation BME Capstone with Gulden Ophthalmics

- Developed a React Native application for concussion evaluation through eye movement tracking

- Integrated react-native-vision-camera and TensorFlow models to automate eye movement recording, and AWS S3 for backend analysis using MediaPipe face tracking landmarks
- https://github.com/ms358/MoVES-MLUI-Mirror

EDUCATION

Rutgers University — Newark, NJ
M.S. Information Technology and Analytics — May 2026
GPA: 3.80/4.00

New Jersey Institute of Technology — Newark, NJ
B.S. Biomedical Engineering — May 2025
GPA: 3.55/4.00 — Cum Laude

SKILLS

Programming: SQL, YAML, Qlik, MATLAB, Python (Scikit-Learn, Pandas, Keras, NumPy), Java, C++
Data Analysis & Visualization: Power BI, PowerQuery, Tableau, Lightdash, Excel (advanced charting, data manipulation, pivot tables, vlookup), MySQL, Sisense, Microsoft Access, Statistical Analysis, MATLAB (EEGLAB, statistical toolboxes)
Software & Tools: JIRA, AzureDevOps, Kanban, Confluence, Visual Studio Code, GitHub, dbt, PostgreSQL, erwin Data Modeler, Microsoft Office Suite, MS Project, Jupyter Notebook, Amazon Redshift, Databricks, AWS (S3), GCP, Linux, VirtualBox, TensorFlow, MediaPipe, SIMULINK, ANSYS, NI Labview, ProtoKinetics Movement Analysis Software